

Nate Cirino

832-833-5202 | cirino.na@northeastern.edu | [portfolio](#) | [linkedin.com/in/nate-cirino](https://www.linkedin.com/in/nate-cirino) | github.com/nateci

Available May 2027 | Open to relocation

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Computer Systems, Algorithms & Data Structures, Computer Networks, Digital Design & Computer Architecture, Systems Security, Cybersecurity, Object-Oriented Design, Foundations of Software Engineering, Foundations of Data Science

SYSTEMS PROJECTS

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft leader election, log replication, and dynamic cluster membership changes across nodes
- Built the durable storage path end to end: memtable, write-ahead log, and leveled on-disk SSTables
- Added MVCC transaction layer providing point-in-time snapshot isolation and linearizable cluster-wide reads
- Designed single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker*

Apr. 2026 – Present

- Built an AI agent autonomously detecting and triaging production incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Designing a supplier-agnostic e-label API unifying 4 vendor integrations behind one interface, powering new outlet launches
- Own front-of-house device stack across all Wayfair physical retail brands (AllModern, Birch Lane, Joss & Main, Perigold): e-labels, mobile, and customer/associate-facing POS
- Shipped POS checkout flows with cross-functional product and retail partners: in-store scan-to-basket customer switching and live associate-entry customer sync

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Debugged and overhauled Relay JMS broker serialization with json-io across 20+ distributed microservices, eliminating message timeouts at enterprise scale
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Optimized Expert AI search and filtering across 20+ microservices, improving clinical query relevance at scale
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Maintained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Architected Flask backend with RAG pipeline for personalized workout recommendations using Bedrock KB
- Evolved RAG stack from LangChain to Vespa to zero-ops Bedrock KB, improving retrieval and reducing overhead
- Led eng org restructure at \$1M-funded startup, mentoring and partitioning teams into DevOps, dev, and agentic workflows

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

SKILLS & HONORS

Languages: C, C++, Python, Java, Rust, Go (learning), TypeScript, JavaScript, Bash, SQL

Systems & Cloud: Linux, Distributed Systems, Kubernetes, Docker, AWS, Azure, Terraform, Jenkins, Buildkite, Groovy, PostgreSQL, GraphQL, Datadog

Frameworks & Libraries: Java Spring, Flask, FastAPI, Tokio, Node.js, Express, React, Next.js, PyTorch, TensorFlow, Pandas

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor